

Sanampalli PavanKalyan

+91-9666310541 — sanampallipavankalyan36@gmail.com

linkedin.com/in/sanampalli-pavankalyan — github.com/SanampalliPavanKalyan04

Professional Summary

Artificial Intelligence and Data Science graduate with a strong foundation in Python programming, machine learning, deep learning, and computer vision. Hands-on experience building AI solutions using TensorFlow, PyTorch, Scikit-learn, CNNs, transfer learning, and ResNet50, along with data preprocessing and model evaluation using Pandas and NumPy. Developed and deployed a computer vision application for plant disease classification using deep learning and Streamlit. Strong problem-solving and analytical skills with a focus on developing practical, data-driven AI solutions. Seeking an entry-level AI/ML Engineer role to apply machine learning and deep learning skills to real-world applications.

Technical Skills

Programming : Python, SQL, Java.

AI/ML : Tensorflow, Pytorch, Scikit-learn, CNN, Transfer Learning, Classification, Computer Vision, Model Deployment.

Data Science : Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing.

Data Visualization : Matplotlib, Power BI, Tableau.

Tools : Jupyter Notebook, Google Colab, VS Code, Git, GitHub.

Internship Experience

Codec Technologies - Data Science Intern

Aug 2025 – Oct 2025

Python, Pandas, NumPy, Matplotlib, Power BI

- I analyzed historical sales to identify key revenue drivers and optimize inventory planning.
- Using Python, Pandas and NumPy, I cleaned and processed raw sales records, then used Matplotlib and Power BI to perform Exploratory Data Analysis (EDA).
- One of my key technical findings was Performed Pareto analysis and discovered that the top 10% of products generated 45% of total revenue.
- I also conducted Year over Year (YoY) growth analysis, which highlighted clear demand spikes during festival and offer periods.
- Using these insights, delivered interactive Power BI dashboards that helped stakeholders forecast demand more accurately and reduce stockouts.

Projects

Plant Disease Detection and Classification Using Machine Learning

Dec 2025

Python, Tensorflow, ResNet50, CNN, Streamlit

- Built an end-to-end computer vision tool to help farmers identify crop diseases early and receive actionable treatment recommendations in real time.
- Trained on 70,000+ images across 38 disease categories from the PlantVillage dataset, applying OpenCV pipelines (rotations, zooms, flips) to prevent overfitting and handle varied real-world lighting.
- Used transfer learning with a fine-tuned ResNet50-based CNN, achieving 95% accuracy and a 0.97 weighted F1-score, prioritizing low false-negative rate for early-stage infections.
- Applied post-training quantization to reduce the model by 70%, drastically reducing latency and memory footprint so it can run efficiently on low-cost edge and IoT devices in the field.
- Built a production-ready API using FastAPI with a responsive UI (Streamlit prototype and React frontend), validating usability and getting feedback from 50+ agricultural stakeholders.
- <https://github.com/SanampalliPavanKalyan04/Plant-disease-detection.git>

Education

Siddhartha Institute of Science and Technology, Puttur, Andhra Pradesh

Oct 2022 – May 2026

Bachelor of Technology in Computer Science Engineering (AI & Data Science)

CGPA: 8.2/10.0

Certifications

- Google AI Essentials – Coursera
- Python for Data Analysis: Pandas & NumPy – Coursera
- What is Generative AI – LinkedIn Learning
- Fundamentals of Machine Learning and Artificial Intelligence – AWS Training
- Data Science Internship – Codec Technologies
- Power BI Data Modelling – Simplilearn
- Industry 4.0 & IIoT – NPTEL
- Claude 101 – Anthropic